AGT 114: PRINCIPLES OF ANIMAL PRODUCTION

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

EDO STATE COLLEGE OF AGRICULTURE AND NATURAL RESOURCES, IGUORIAKHI CAMPUS

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AGT 111: PRINCIPLES OF ANIMAL PRODUCTION

Number of units: 3 Units

60 Hours (2 Hours Lectures, 2 Hours Practical)

4hours / week

Course synopsis

- Classification of livestock animals
- Management of animal production
- Classification of feedstuffs
- Ruminant animals:
    - Cattle:
        - Breeding and reproduction in cattle
        - Management of beef and dairy cattle
    - Sheep and goat
- Non-ruminant/ Monogastric animals:
    - Breeding and reproduction in swine
    - Poultry breeds and types of poultry
    - Management of layers and broilers
    - Rabbit and cane rat (grasscutter): care and management
    - Livestock diseases and parasites

GENERAL OBJECTIVES:
On completion of this course, the student should be able to:

1. **Appreciate contribution of livestock to the national economy and areas of major livestock production in Nigeria**
    - **a.** Explain social and economic uses of farm animals
    - **b.** Explain the geographical distribution of livestock production zones
    - **c.** Explain the supply chain for animal products and indicate how important it is for the Nigerian
2. **Appreciate different livestock rearing systems and their advantages and disadvantages**
    - **a.** Describe:
        - **i.** Extensive livestock management systems
        - **ii.** Semi extensive livestock management systems
        - **iii.** Intensive livestock management systems
    - **b.** Explain the advantages and disadvantages of each system
3. **Understand different classes of feed resources and processing techniques**
    - **a.** Guide students to identify and classify feedstuff into carbohydrate, protein, lipid and fibre sources.
    - **b.** Explain the difference between roughages, concentrates etc.
    - **c.** Explain the importance of minerals and vitamins in animal rations.
    - **d.** Explain the need for processing and how to process
4. **Understand monogastric, ruminant and pseudo-ruminant digestive systems**
    - **a.** Understand the physiology of digestive systems of monogastric, ruminant and pseudo ruminant animals

   b. Understand the biochemistry of digestive systems of monogastric, ruminant and pseudo ruminant animals
   c. Understand how to calculate a simple ration formulation by hand
5. **Appreciate different nutrients and their role in animal growth and production**
   a. Introduce the biochemical characterization of each group of feedstuffs and explain their pathway inside the animal.
6. **Understand reproductive physiology and genetic improvement of farm animals.**
   a. Understand the influence of hormones on male and female reproduction.
   b. Know the physiology of the male and female reproductive organs.
   c. Understand what happens at fertilization, gestation and parturition.
   d. Understand the concepts of animal genetics and how breed improvement can be managed
7. **Appreciate basic farm animals' health management principles**
   a. Explain how diseases affect animals and how they are spread.
   b. Explain what parasites are and how they interact with animals
   c. Explain the concepts of preventative medicine, flock or herd health, clean grazing techniques etc

COMMON TERMS IN ANIMAL PRODUCTION

A. General Terms (Production Basics)

1. **Livestock** – Domesticated farm animals raised for food, fiber, or labor.
2. **Poultry** – Domesticated birds such as chickens, turkeys, ducks, and quails raised for meat and eggs.
3. **Breed** – A group of animals with similar characteristics passed on to their offspring.
4. **Species** – A group of animals capable of interbreeding (e.g., cattle, sheep, goats).
5. **Ruminant** – Animals with a four-chambered stomach (e.g., cattle, sheep, goats).
6. **Monogastric** – Animals with a single-chambered stomach (e.g., pigs, poultry, rabbits).
7. **Weaning** – The transition of a young animal from milk to solid feed.
8. **Breeding** – The act of mating animals to produce offspring.
9. **Gestation** – The period between conception and birth.
10. **Parturition** – The act of giving birth.

B. Nutrition and Feeding

11. **Ration** – The amount of feed given to an animal daily.
12. **Balanced Diet** – A feed mixture that contains all essential nutrients in the correct proportions.
13. **Feed Conversion Ratio (FCR)** – The amount of feed required to gain a unit of body weight.
14. **Silage** – Fermented, high-moisture fodder preserved for ruminant feed.
15. **Hay** – Dried grasses and legumes used as fodder.
16. **Supplement** – A feed added to improve the nutrient value of the base diet.
17. **Ad libitum feeding** – Allowing animals to eat as much as they want.
18. **Roughages** – High-fiber feeds like hay and straw.
19. **Concentrates** – Low-fiber, high-energy feeds (e.g., grains, oilseeds).
20. **Mineral Lick** – A block or mixture containing essential minerals.

C. Health and Welfare

21. **Vaccination** – Administering vaccines to protect animals against diseases.
22. **Deworming –** Treating animals to remove internal parasites.
23. **Quarantine –** Isolating new or sick animals to prevent disease spread.
24. **Biosecurity** – Measures taken to reduce disease risk in animal production.
25. **Animal Welfare** – Ensuring the humane treatment of animals.
26. **Veterinary Care** – Health care provided by animal health professionals.
27. **Zoonotic Diseases –** Diseases that can be transmitted from animals to humans.
28. **Morbidity** – The rate of illness in a group of animals.
29. **Mortality** – The rate of death in a group of animals.
30. **Culling** – Removing underperforming or sick animals from the herd/flock.

D. Reproduction and Genetics

31. **Artificial Insemination (AI)** – Fertilization using collected semen instead of natural mating.
32. **Estrus (Heat)** – The period when a female is receptive to mating.
33. **Oestrus Cycle** – The reproductive cycle in female mammals.
34. **Ovulation** – The release of an egg from the ovary.
35. **Lactation –** The production of milk by the mammary glands.
36. **Pedigree** – A record of an animal's ancestry.
37. **Selection –** Choosing animals with desirable traits for breeding.
38. **Crossbreeding** – Mating animals from different breeds.
39. **Heterosis (Hybrid Vigor)** – Improved performance of crossbred offspring.
40. **Inbreeding –** Mating closely related animals.

E. Housing, Management, and Production Systems

41. **Housing** – Structures provided for sheltering animals.
42. **Stocking Density** – The number of animals per unit area.
43. **Extensive System** – Low-input, large-area farming (e.g., grazing).
44. **Intensive System** – High-input, confined animal production system.
45. **Semi-intensive System** – Combines grazing with supplemental feeding.
46. **Record Keeping –** Documentation of animal performance, health, and inputs.
47. **Carcass Yield** – The usable meat obtained from a slaughtered animal.
48. **Market Weight** – The weight at which an animal is sold for slaughter.
49. **Ethology** – The scientific study of animal behavior.
50. **Traceability** – The ability to track an animal or its products through all stages of production and distribution.

PHYSIOLOGY OF THE DIGESTIVE SYSTEMS OF MONOGASTRIC, RUMINANT, AND PSEUDO-RUMINANT ANIMALS

The digestive system plays a vital role in animal physiology by breaking down ingested food into absorbable nutrients. However, the anatomy and physiology of the digestive tract vary significantly among animal species depending on their feeding habits and evolutionary adaptations. Animals can be broadly categorized into three digestive types: monogastric, ruminant, and pseudo-ruminant. Each type presents unique structural and functional modifications to support their specific dietary needs.

MONOGASTRIC ANIMALS

Monogastric animals possess a simple, single-chambered stomach. Examples include humans, pigs, dogs, poultry, and horses (although horses exhibit some hindgut fermentation traits, they are still largely monogastric). These animals primarily consume diets rich in concentrated nutrients, such as carbohydrates, proteins, and fats, and rely largely on enzymatic digestion.

Anatomy and Physiology: The monogastric digestive system begins at the mouth, where mechanical digestion via mastication (chewing) and chemical digestion via salivary enzymes (e.g., amylase in pigs) start breaking down starch. Food then travels down the esophagus to the stomach.

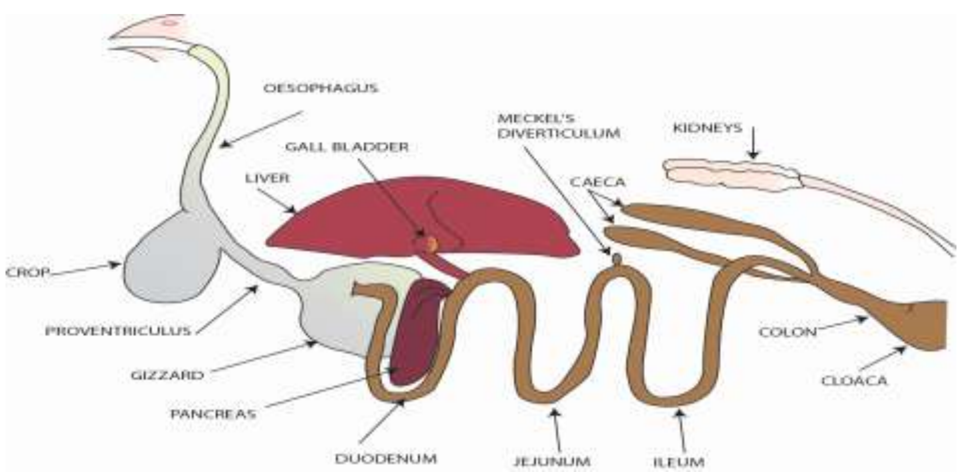
In the **stomach**, hydrochloric acid (HCl) and pepsin initiate protein breakdown. The stomach also acts as a temporary food reservoir and mixes its contents into chyme. From the stomach, chyme passes into the **small intestine**, which is the primary site of enzymatic digestion and nutrient absorption. Digestive enzymes secreted by the pancreas and bile from the liver aid in breaking down fats, proteins, and carbohydrates.

The **duodenum**, **jejunum**, and **ileum** are the three segments of the small intestine. Absorption occurs mostly in the jejunum and ileum through villi and microvilli that increase the surface area for nutrient uptake into the bloodstream.

Undigested material proceeds to the **large intestine**, where water, electrolytes, and some volatile fatty acids (VFAs) from limited microbial fermentation (especially in pigs) are absorbed. The remaining waste is compacted into faeces and expelled.

Special Features: Monogastric animals are generally less efficient at digesting fibrous materials due to the absence of extensive microbial fermentation chambers. Their diets typically require higher-quality feed.

THE DIGESTIVE SYSTEM OF A CHICKEN



RUMINANT ANIMALS

Ruminants, such as cattle, sheep, goats, deer, and buffaloes, have a complex four-compartment stomach that allows them to efficiently digest fibrous (forage and grasses) plant material through microbial fermentation. These animals are adapted to herbivorous diets rich in cellulose and hemicellulose.

Stomach Compartments:

1. **Rumen**: This is the largest compartment and serves as a fermentation vat. It hosts a diverse population of microbes (bacteria, protozoa, fungi) that break down fibrous plant materials into volatile fatty acids (VFAs), methane, and microbial proteins. VFAs (primarily acetate, propionate, and butyrate) are the major energy source for ruminants.
2. **Reticulum**: Often considered an extension of the rumen, the reticulum has a honeycomb-like structure and aids in trapping dense objects (hardware disease prevention). It also plays a crucial role in regurgitating cud for rumination (re-chewing), which further reduces particle size for better microbial action.
3. **Omasum**: This compartment functions mainly in water and electrolyte absorption. Its many folds increase the surface area for absorption.
4. **Abomasum**: Known as the "true stomach," the abomasum resembles the monogastric stomach. It secretes hydrochloric acid and digestive enzymes (e.g., pepsin) to digest microbial protein and any bypass dietary protein.

Post-Stomach Digestion: After passing through the abomasum, the digesta enters the **small intestine**, where further digestion and absorption of nutrients, especially amino acids, fatty acids, glucose (from propionate conversion), and vitamins occur.

Large Intestine Function: The large intestine, like in monogastrics, functions in water absorption and minor fermentation of remaining undigested materials. However, the nutrients produced here are less available for absorption.

Special Features: Ruminants rely heavily on microbial symbiosis to utilize fibrous feeds. The microorganisms synthesize essential amino acids and B-complex vitamins, allowing ruminants to thrive on low-quality forages.

Digestive system of cattle

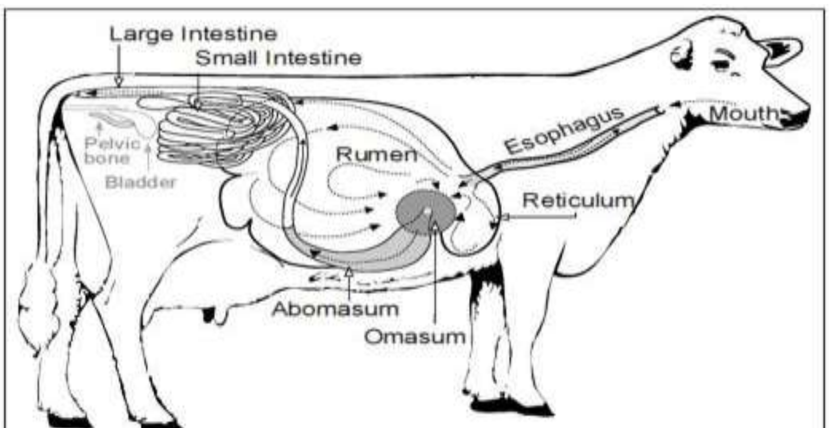


Figure 1: The digestive system of a cow includes four stomachs. The rumen is similar to a lake with a river running through one corner.

PSEUDO-RUMINANT ANIMALS

Pseudo-ruminants, such as camels, llamas, alpacas, and hippopotamuses, share similarities with true ruminants but possess distinct differences in their digestive physiology. They are primarily herbivores with an efficient fermentation system, although their stomach structure is not the same as ruminants.

Stomach Compartments: Pseudo-ruminants have a **three-compartment stomach**, typically lacking the omasum:

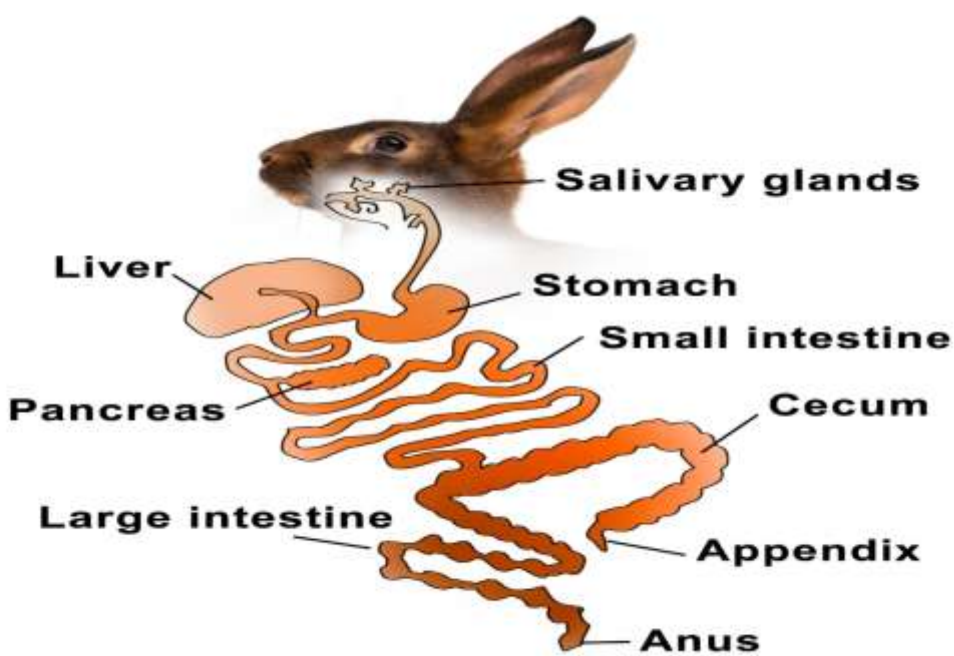
1. **C1 (analogous to the rumen)**: This large compartment functions as a fermentation chamber. It hosts microbes similar to the rumen and aids in the breakdown of cellulose into VFAs.
2. **C2 (analogous to the reticulum)**: Aids in particle sorting and further fermentation. It also initiates the cud-chewing process.
3. **C3 (partially analogous to the abomasum)**: This is the tubular compartment where acid and enzymatic digestion take place. The distal part of C3 produces HCl and pepsin for protein digestion.

Digestive Efficiency: Despite not having a true rumen or omasum, pseudo-ruminants are efficient fiber digesters. They exhibit regurgitation and rumination behaviors similar to true ruminants and rely on microbial fermentation for energy.

Intestinal Function: After leaving C3, the digesta proceeds to the **small intestine**, where nutrient absorption occurs, followed by water and mineral absorption in the **large intestine**.

Special Features: Camels, for instance, have adapted to arid environments by having highly efficient water reabsorption mechanisms and can digest thorny, low-quality vegetation.

Digestive system of a rabbit



Comparative Overview

Feature	Monogastric	Ruminant	Pseudo-Ruminant
Stomach Chambers	One	Four (Rumen, Reticulum, Omasum, Abomasum)	Three (C1, C2, C3)
Fiber Digestion	Poor	Excellent (microbial fermentation)	Good (microbial fermentation)
Microbial Fermentation	Limited (hindgut, e.g., pigs)	Extensive (foregut)	Moderate to extensive (foregut)
Main Energy Source	Glucose	Volatile Fatty Acids (VFAs)	VFAs
Cud-Chewing	Absent	Present	Present
Enzyme Production	Begins in mouth and stomach	Begins mainly post-rumination	Begins in C3 compartment

BIOCHEMISTRY OF THE DIGESTIVE SYSTEMS OF MONOGASTRIC, RUMINANT, AND PSEUDO-RUMINANT ANIMALS

Digestion is a complex biochemical process involving the breakdown of macronutrients—carbohydrates, proteins, and lipids into absorbable units. This process is facilitated by enzymes, acids, and microbial metabolites. Different animal species have evolved distinct biochemical pathways in their digestive systems based on their diets and gut morphology.

The three primary digestive categories—**monogastric, ruminant, and pseudo-ruminant** exemplify unique biochemical adaptations that influence how nutrients are processed and absorbed.

1. Digestion processes in Monogastric Animals

Monogastric animals such as pigs, poultry, and humans possess a single-chambered stomach. Their digestive biochemistry primarily relies on **endogenous enzymatic activity** rather than microbial fermentation.

Biochemical Processes:

- **Carbohydrate Digestion:** Begins in the mouth (in some species) with **salivary amylase**, which hydrolyzes starch to maltose. The major digestion occurs in the small intestine through **pancreatic amylase**, converting starch to maltose, which is further broken down by **maltase, sucrase, and lactase** at the brush border into glucose, fructose, and galactose for absorption.
- **Protein Digestion:** Initiated in the stomach by **pepsin**, activated by hydrochloric acid. Further breakdown occurs in the small intestine with **pancreatic enzymes**—trypsin, chymotrypsin, and carboxypeptidase converting polypeptides to amino acids and small peptides.
- **Lipid Digestion:** Starts in the small intestine with **bile salts** emulsifying fats, followed by action of **pancreatic lipase** that hydrolyzes triglycerides into monoglycerides and free fatty acids. These form micelles for absorption across the intestinal epithelium (wall).
- **Vitamin and Mineral Absorption:** Water-soluble vitamins (B-complex, C) and fat-soluble vitamins (A, D, E, K) are absorbed mainly in the small intestine. Iron, calcium, and magnesium are absorbed in their ionized forms.
- **Minimal Fermentation:** Some fermentation by colonic microbiota in the large intestine produces small amounts of **short-chain fatty acids (SCFAs)** like acetate and butyrate.

2. Digestion processes in Ruminant Animals

Ruminants such as cattle, sheep, and goats exhibit highly adapted **microbial fermentation** biochemistry, particularly in the rumen. They depend heavily on **symbiotic microorganisms** to break down complex plant polysaccharides.

Biochemical Processes:

- **Fermentation in the Rumen:**
    - **Cellulose and hemicellulose** are fermented by microbes to **volatile fatty acids (VFAs)**—primarily **acetate, propionate, and butyrate** which are absorbed through the rumen wall and serve as the main energy source, thereby converting hydrogen and carbon dioxide into methane (CH_4), which is a gas passed through the anal orifice.
- **Protein Metabolism:**
    - **Dietary proteins** are degraded by microbes into **ammonia, peptides, and amino acids**. Ammonia is utilized by microbes to synthesize **microbial protein**, which is later digested in the abomasum and small intestine.
    - **Non-protein nitrogen (NPN)**, like urea, is recycled into the rumen via saliva or the rumen wall, allowing microbial protein synthesis.
- **Lipids:**
    - **Lipolysis** in the rumen converts dietary triglycerides into glycerol and free fatty acids. Glycerol is fermented to VFAs.
- **Vitamin Synthesis:**
    - Microbes synthesize **B-vitamins** and **vitamin K**, eliminating the need for dietary supplementation.
- **Post-Rumen Digestion:**
    - In the **abomasum**, pepsin and HCl break down microbial and bypass proteins.
    - In the **small intestine**, enzymatic digestion and nutrient absorption mirror that of monogastrics.

3. Digestion processes in Pseudo-Ruminant Animals

Pseudo-ruminants, such as camels and llamas, have a **three-chambered stomach** and rely on **foregut fermentation**, though less compartmentalized than ruminants. Their digestive biochemistry is a hybrid between monogastric and ruminant systems.

Biochemical Processes:

- **Fermentation (Foregut):**
  - Occurs mainly in **compartments C1 and C2**, where **VFAs** are produced by microbial fermentation of fibrous material.
  - Like ruminants, they absorb acetate, propionate, and butyrate for energy.
- **Protein Utilization:**
  - Dietary and microbial proteins are digested in **C3**, where **HCl and pepsin** act similarly to the monogastric stomach.
  - Urea recycling also occurs, supporting microbial protein synthesis.
- **Lipid Metabolism:**
  - Lipids are hydrolyzed and biohydrogenated in the fermentation chambers, though less extensively than in ruminants.
- **Vitamin Production:**
  - B-vitamins and vitamin K are synthesized by gut microbes, reducing dietary dependency.
- **Small Intestine Activity:**
  - Enzymatic digestion of protein, carbohydrates, and lipids occurs with pancreatic secretions and brush-border enzymes.

Comparative Biochemical Overview

Parameter	**Monogastric**	**Ruminant**	**Pseudo-Ruminant**
Stomach Chambers	1	4 (Rumen, Reticulum, Omasum, Abomasum)	3 (C1, C2, C3)
Main Digestion Type	Enzymatic	Microbial + Enzymatic	Microbial + Enzymatic
Carbohydrate End Product	Glucose	VFAs (Acetate, Propionate, Butyrate)	VFAs
Main Energy Source	Glucose	VFAs	VFAs
Protein Source	Dietary protein	Microbial protein + Bypass protein	Microbial protein + Dietary protein
Fat Digestion Site	Small intestine (pancreatic lipase)	Rumen (biohydrogenation), Small intestine	Foregut + Small intestine
Vitamin Synthesis	Minimal (some by colon microbes)	Extensive (B-vitamins, vitamin K by rumen microbes)	Moderate (B-vitamins, vitamin K by foregut microbes)
Urea Recycling	Minimal	Active (via saliva and rumen wall)	Active
Methane Production	Negligible	High (via archaea in rumen)	Moderate
Fiber Utilization	Poor	Excellent	Good

The biochemical processes underlying digestion in monogastric, ruminant, and pseudo-ruminant animals reflect significant evolutionary adaptations to their diets. Monogastrics depend on endogenous enzymes for nutrient breakdown, making them efficient at digesting simple carbohydrates and proteins but less effective at processing fiber. Ruminants possess a unique symbiotic microbial ecosystem in their rumen, enabling the fermentation of fibrous plant materials into VFAs and the synthesis of microbial protein and vitamins. Pseudo-ruminants occupy a middle ground, utilizing foregut fermentation efficiently, though their compartmentalization and biochemical processes are simpler than those of true ruminants.

These distinctions are crucial in animal nutrition and health management, especially in designing species-appropriate feeding strategies, enhancing feed efficiency, and reducing digestive-related disorders. Understanding digestive biochemistry not only informs livestock production practices but also aids in improving nutrient utilization, reducing environmental emissions, and maximizing productivity across animal species.

CALCULATING SIMPLE FEED RATION FORMULATION BY HAND

Feed ration formulation is a critical process in animal nutrition, ensuring that livestock receive the appropriate nutrients for maintenance, growth, reproduction, or production. A simple feed formulation by hand involves basic mathematical calculations using known nutrient values and requirements. One of the most common methods used is the **Pearson Square Method**. This technique helps to determine the proportion of two feed ingredients needed to meet a specific nutrient requirement, typically crude protein.

Steps in Using the Pearson Square Method:

1. **Draw a square.**
2. **Place the target nutrient level (e.g., 16% crude protein) in the center.**
3. **List the nutrient levels of the two feed ingredients on the left corners of the square (top and bottom).**
4. **Subtract diagonally (larger minus smaller) to find the parts needed of each feed.**
5. **Sum the parts to get the total mixture.**
6. **Calculate the percentage of each ingredient in the ration.**

Example 1:

Formulate a 16% CP ration using maize (8.5% CP) and soybean meal (44% CP).

matlab
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```
 Maize (8.5%)
     |\
     | \  = 28.0 parts SBM
16% CP
     | /  = 7.5 parts Maize
 SBM (44%)
```

Total = 28 + 7.5 = 35.5 parts

% Maize = (7.5 / 35.5) x 100 = 21.1%
% SBM = (28 / 35.5) x 100 = 78.9%

Example 2:

Target: 18% CP; Ingredients: Wheat offal (15% CP) and Groundnut cake (40% CP)

Result: 88% wheat offal and 12% groundnut cake.

Example 3:

Target: 20% CP; Ingredients: Cassava peel (5% CP) and Fish meal (60% CP)

Result: 8.0% cassava peel and 92.0% fish meal.

Example 4:

Target: 12% CP; Ingredients: Corn bran (10% CP) and Sunflower meal (35% CP)

Result: 92% corn bran and 8% sunflower meal.

Example 5:

Target: 14% CP; Ingredients: Sorghum (9% CP) and Cottonseed meal (41% CP)

Result: 84% sorghum and 16% cottonseed meal.

Example 6:

Target: 22% CP; Ingredients: Rice bran (13% CP) and Blood meal (80% CP)

Result: 89% rice bran and 11% blood meal.

BIOCHEMICAL PATHWAYS OF CARBOHYDRATES, LIPIDS, AND PROTEINS

1. Carbohydrate Metabolism

Carbohydrates are the primary energy source for most animals. Their metabolism involves:

Digestion and Absorption

- Begins in the **mouth** (salivary amylase) and continues in the **small intestine** (pancreatic amylase).
- Polysaccharides → disaccharides → monosaccharides (mainly glucose).
- Glucose is absorbed into the bloodstream via **active transport**.

Biochemical Pathways

- **Glycolysis**: Occurs in the cytoplasm; glucose (6C) is broken into 2 pyruvate (3C) molecules, producing **ATP** and **NADH**.
- **Krebs Cycle (Citric Acid Cycle)**: In the mitochondria, pyruvate is converted to acetyl-CoA, which enters the cycle to produce more **NADH**, **$FADH_2$**, and **CO_2**.
- **Electron Transport Chain (ETC)**: Uses NADH and $FADH_2$ to generate a large amount of ATP through **oxidative phosphorylation**.
- **Gluconeogenesis**: Formation of glucose from non-carbohydrate sources (e.g., amino acids) during fasting.

2. Lipid Metabolism

Lipids are dense energy sources and are essential for cell membranes and hormone synthesis.

Digestion and Absorption

- Begins in the **small intestine** with bile salts emulsifying fats.
- Lipases break triglycerides into **fatty acids** and **monoglycerides**, which are absorbed via **micelles** into enterocytes.

Biochemical Pathways

- **β-Oxidation**: In mitochondria, fatty acids are broken into acetyl-CoA units, which enter the **Krebs cycle**.
- **Ketogenesis**: In the liver, excess acetyl-CoA from fatty acids can form **ketone bodies**, an alternative energy source.
- **Lipogenesis**: In the liver and adipose tissue, excess glucose is converted into **fatty acids** and stored as triglycerides.

3. Protein Metabolism

Proteins are not primarily energy sources but are essential for tissue repair, enzymes, hormones, and immune function.

Digestion and Absorption

- Begins in the **stomach** (pepsin) and continues in the **small intestine** (trypsin, chymotrypsin).
- Proteins → peptides → amino acids, absorbed via active transport into the bloodstream.

Biochemical Pathways

- **Deamination**: Removal of the amino group in the liver, producing **ammonia**, which is converted to **urea** and excreted.
- **Transamination**: Amino groups transferred to keto acids to synthesize non-essential amino acids.
- **Amino Acid Catabolism**: Carbon skeletons enter **glycolysis** or **Krebs cycle** for energy or gluconeogenesis.

PATHWAY SUMMARY

Nutrient	Key Pathways	End Products	Energy Yield
Carbohydrates	Glycolysis, Krebs, ETC	ATP, CO_2, H_2O	High (quick)
Lipids	β-oxidation, Krebs, ETC	ATP, CO_2, H_2O, Ketones	Very High
Proteins	Deamination, Transamination, Krebs	Urea, ATP	Moderate

UNDERSTAND THE INFLUENCE OF HORMONES ON MALE AND FEMALE REPRODUCTION

Reproduction: Reproduction is a fundamental biological process of animal giving birth to species of its kind.

Parturition: This is the process giving birth in the animal

In mammals, this process is intricately regulated by a series of hormonal interactions that govern the development, function, and coordination of the male and female reproductive systems. These hormonal controls ensure the proper timing of events such as gametogenesis, mating behavior, fertilization, gestation, and parturition.

I. Endocrine Control of Reproduction

The **hypothalamic-pituitary-gonadal (HPG) axis** is the central endocrine system that regulates reproductive function in both males and females.

A. Hormonal Influence in Males

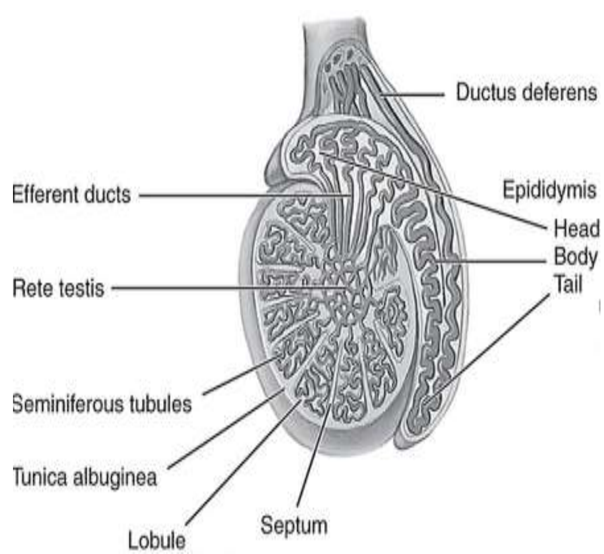
1. **Gonadotropin-Releasing Hormone (GnRH):**
   - Secreted by the **hypothalamus** in a pulsatile manner.
   - Stimulates the **anterior pituitary gland** to release **Luteinizing Hormone (LH)** and **Follicle Stimulating Hormone (FSH)**.
2. **Luteinizing Hormone (LH):**
   - Targets the **Leydig cells** in the testes.
   - Stimulates production of **testosterone**, the principal male sex hormone.
3. **Testosterone:**
   - Essential for the development of **male secondary sexual characteristics** (e.g., deep voice, muscle mass, facial hair).
   - Maintains **sperm production**, **libido**, and **reproductive organ integrity**.
   - Provides **negative feedback** to the hypothalamus and pituitary to regulate GnRH and LH secretion.

B. Hormonal Influence in Females

1. **GnRH:**
   - Released by the hypothalamus, stimulates the anterior pituitary to produce LH and FSH.
2. **FSH:**
   - Stimulates the growth and maturation of **ovarian follicles**.
   - Follicular cells produce **estrogen** in response.
3. **LH:**
   - Surge in LH leads to **ovulation**.
   - Supports the formation and maintenance of the **corpus luteum**, which produces **progesterone**.
4. **Estrogen:**
   - Produced by **growing follicles**.
   - Stimulates development of **female secondary sexual characteristics** and prepares the endometrium for possible implantation.
   - Low to moderate levels provide **negative feedback** to reduce FSH and LH.
   - High levels trigger **positive feedback**, causing LH surge and ovulation.
5. **Progesterone:**
   - Secreted by the **corpus luteum** post-ovulation.
   - Prepares and maintains the **endometrium** for embryo implantation.
   - Inhibits further ovulation during pregnancy by suppressing GnRH.
6. **Inhibin:**
   - Secreted by granulosa cells of the follicles.
   - Selectively inhibits FSH to regulate follicular development.

II. Physiology of Male Reproductive Organs

STRUCTURE OF A BULL TESTIS



The male reproductive system is designed for **sperm production**, **storage**, and **delivery** to the female reproductive tract.

A. Testes

- Paired oval glands located in the scrotum.
- Contain **seminiferous tubules** for **spermatogenesis**.
- **Leydig cells** produce testosterone.

B. Epididymis

- Coiled tube where sperm mature and are stored.
- Provides motility and fertilizing ability to sperm.

C. Vas Deferens

- Muscular tube transporting mature sperm from the epididymis to the urethra.

D. Accessory Glands

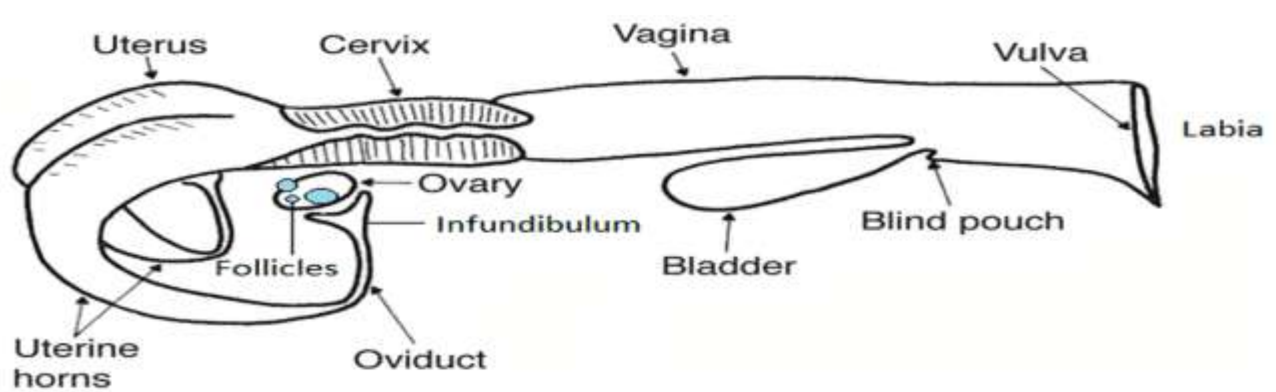
1. **Seminal Vesicles**: Contribute fluid rich in fructose to nourish sperm.
2. **Prostate Gland**: Adds enzymes and alkaline fluids that enhance sperm motility.
3. **Bulbourethral Glands**: Secrete mucus that neutralizes urinary acids in the urethra.

E. Penis

- Contains erectile tissue.
- Facilitates deposition of sperm into the female tract during copulation.

III. Physiology of Female Reproductive Organs

Structure of a cow ovary



The female reproductive system supports **oocyte production**, **fertilization**, and **gestation**.

A. Ovaries

- Paired organs producing **ova (eggs)** and **sex hormones** (estrogen and progesterone).
- Contain **follicles** at various stages of development.
- Responsible for **ovulation** and formation of **corpus luteum**.

B. Oviducts (Fallopian Tubes)

- Tubes with **fimbriae** that capture the ovulated egg.
- Site of **fertilization**.
- Transports the zygote to the uterus.

C. Uterus

- Thick-walled, muscular organ with **endometrial lining**.
- Site for **implantation**, **placentation**, and **fetal development**.
- Contracts during parturition to expel the fetus.

D. Cervix

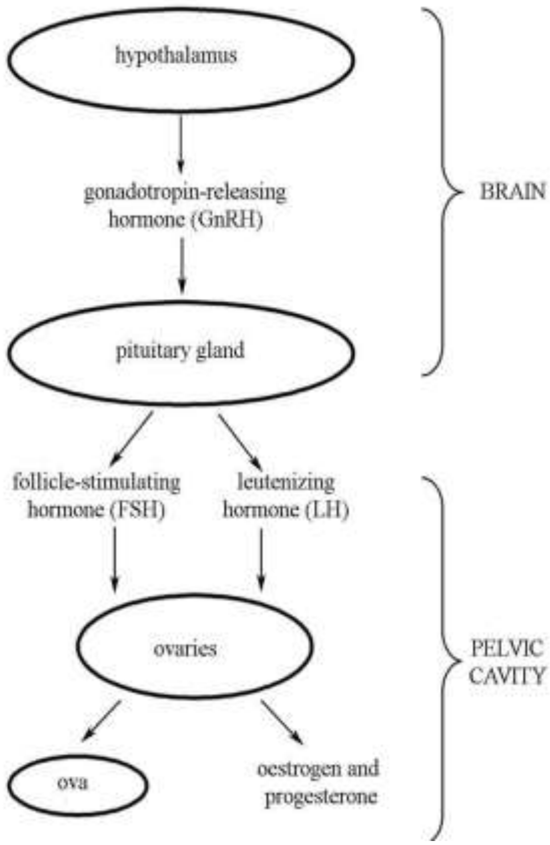
- Narrow outlet of the uterus.
- Acts as a **sperm reservoir** and **barrier to pathogens**.
- Dilates during labor.

E. Vagina

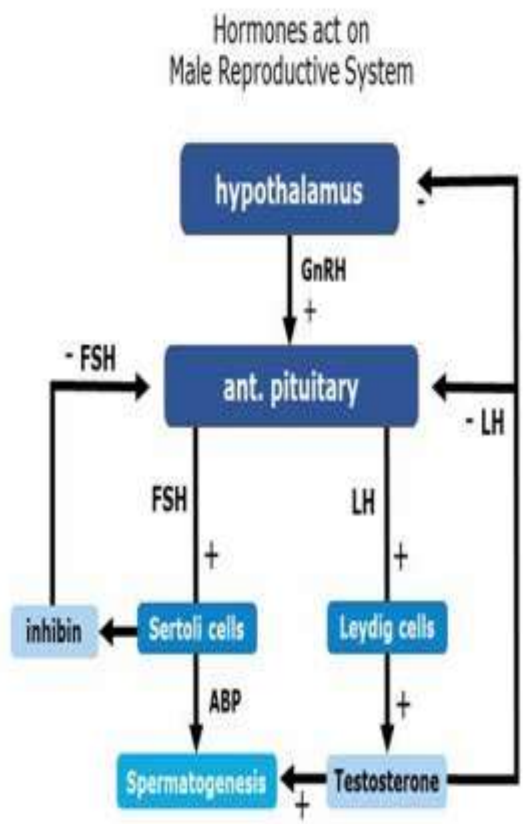
- Muscular canal serving as the **copulatory organ** and **birth canal**.
- Receives sperm during mating.

IV. Diagram: Hormonal Regulation of Reproduction

FEMALE



MALE



V. Table: Comparative Overview of Male and Female Reproductive Systems and Hormones

Parameter	**Male**	**Female**
Primary Gonads	Testes	Ovaries
Key Hormones	Testosterone, FSH, LH	Estrogen, Progesterone, FSH, LH
Supporting Cells	Sertoli and Leydig cells	Granulosa and Theca cells
Gamete	Spermatozoa	Ovum (egg cell)
Site of Gamete Formation	Seminiferous tubules	Ovarian follicles

Parameter	Male	Female
Hormone Source	Leydig cells (testosterone)	Follicular and luteal cells
Hormone Feedback	Testosterone inhibits GnRH, LH	Estrogen and Progesterone inhibit GnRH, FSH, LH
Function of FSH	Stimulates spermatogenesis	Stimulates follicle growth
Function of LH	Testosterone production (Leydig)	Ovulation and corpus luteum
Accessory Organs	Seminal vesicles, prostate	Uterus, cervix, vagina
Purpose of Hormones	Sexual traits, libido, sperm prod.	Menstrual cycle, pregnancy prep

Understanding hormonal influences on reproduction is critical in:

- **Animal breeding and fertility management**.
- **Artificial insemination and synchronization of estrus**.
- **Treatment of reproductive disorders** (e.g., polycystic ovary syndrome, hypogonadism).
- **Birth control and hormone therapy**.

In veterinary and animal science, manipulating these hormonal systems using **hormone injections** (e.g., GnRH analogs, prostaglandins, progestins) is common practice to **synchronize estrus**, improve **conception rates**, or **induce parturition**.

UNDERSTANDING WHAT HAPPENS AT FERTILIZATION, GESTATION, AND PARTURITION: HORMONAL RESPONSES INCLUDED

Reproduction in animals is a complex physiological process involving three critical stages: fertilization**,** gestation**, and** parturition**. These stages are tightly regulated by a series of coordinated** hormonal events **that ensure successful development of the embryo, maintenance of pregnancy, and eventual delivery of offspring.**

1. Fertilization

Fertilization is the union of the male and female gametes—sperm and ovum—resulting in the formation of a **zygote**. This usually occurs in the **oviduct (fallopian tube)** of the female reproductive tract.

- When the sperm enters the female tract, it undergoes **capacitation**, a process that enables it to penetrate the ovum's protective layers.
- Once a single sperm successfully enters the ovum, it triggers a **cortical reaction** that prevents other sperms from fusing, ensuring **monospermy**.

- The male and female genetic materials fuse, forming a diploid zygote with a complete set of chromosomes (half from each parent).

Hormonal Influence:

- **Luteinizing Hormone (LH)** from the anterior pituitary triggers ovulation, releasing the ovum for potential fertilization.
- **Estrogen** and **progesterone** prepare the reproductive tract (especially the uterus) for sperm reception and possible implantation.

2. Gestation

Gestation is the period of embryo and fetal development within the uterus, lasting from fertilization to birth. In mammals, the zygote undergoes cleavage and develops into a **blastocyst**, which implants into the **endometrial lining** of the uterus.

Once implantation occurs, the embryo forms a **placenta**, which serves as the interface for nutrient and gas exchange between mother and fetus. The placenta also becomes an endocrine organ producing key hormones.

Hormonal Responses during Gestation:

- **Progesterone**:
  - Secreted initially by the **corpus luteum** and later by the **placenta**.
  - Maintains the uterine lining and inhibits uterine contractions to prevent miscarriage.
- **Estrogen**:
  - Increases gradually throughout pregnancy.
  - Stimulates uterine growth and enhances blood flow to the placenta.
  - Prepares the mammary glands for lactation.
- **Human Chorionic Gonadotropin (hCG)** (in humans) or similar hormones in animals (e.g., **interferon tau** in cattle):
  - Maintains the corpus luteum during early pregnancy.
  - Prevents the regression of progesterone production.
- **Relaxin**:
  - Produced by the placenta and corpus luteum.
  - Softens pelvic ligaments and cervix to prepare for birth.
- **Placental Lactogen**:
  - Helps regulate maternal metabolism and promotes fetal growth.
  - Prepares the mammary glands for lactation.

TABULAR REPRESENTATION

Animal	Age at Sexual Maturity	Heat Cycle (Estrus Interval)	Ovulation Period	Gestation Period
Poultry (e.g., Chicken)	18–24 weeks (5–6 months)	No true estrus cycle (continuous ovulation)	Ovulation occurs ~30–75 minutes after egg laying	~21 days (egg incubation)
Sheep (Ewe)	5–8 months	16–17 days	24–30 hours after onset of estrus	~147 days (5 months)
Goat (Doe)	5–8 months	18–21 days	24–36 hours after onset of estrus	~150 days (5 months)
Pig (Sow)	5–7 months	18–24 days	36–44 hours after onset of estrus	~114 days (3 months, 3 weeks, 3 days)
Cattle (Cow)	8–12 months (breed dependent)	18–24 days	~12 hours after end of estrus	~283 days (9 months)
Rabbit (Doe)	4–6 months	Induced ovulator (no cycle)	10–13 hours after mating	~28–32 days (1 month)

3. Parturition

Parturition is the act of giving birth, involving the expulsion of the fetus and placenta from the uterus through the birth canal. This stage is initiated by a **complex neuroendocrine cascade**, usually triggered by the **maturing fetus**.

Hormonal Changes During Parturition:

- **Fetal Cortisol**:
  - As the fetus matures, its adrenal glands release **cortisol**, which acts on the placenta to shift hormone production.
  - This results in a **decrease in progesterone** and a **rise in estrogen** levels.
- **Estrogen**:
  - Increases **uterine sensitivity to oxytocin**.
  - Promotes the production of **prostaglandins** (especially PGF2α), which help in cervical dilation and uterine contractions.
- **Oxytocin**:
  - Released by the **posterior pituitary gland** in response to cervical stretching.
  - Stimulates **strong uterine contractions** (positive feedback loop).
  - Also plays a role in milk ejection during lactation.
- **Prostaglandins (PGF2α)**:
  - Aid in uterine contractions and **lysis of the corpus luteum**, stopping progesterone production.
  - Facilitate cervical ripening and the separation of fetal membranes.

The **stages of parturition** include:

1. **Dilation of the cervix** (latent phase),
2. **Expulsion of the fetus** (active phase),
3. **Expulsion of the placenta** (afterbirth).

CONCEPTS OF ANIMAL GENETICS AND BREED IMPROVEMENT

Animal genetics: This is the branch of science that studies heredity and variation in animals. It involves understanding how genetic traits are passed from parents to offspring and how these traits influence performance characteristics such as growth rate, milk production, disease resistance, fertility, and behavior. Animal genetics forms the basis for breed improvement, which aims to enhance desirable traits in farm animals to increase productivity and efficiency.

At the core of animal genetics are **genes**, which are segments of DNA that determine specific traits. These genes are inherited according to Mendelian laws, where dominant and recessive alleles influence the phenotype (observable traits). By identifying and selecting animals with superior genetics, farmers and breeders can achieve progressive improvements in animal performance over generations.

Breed Improvement Methods

Breed improvement refers to the process of enhancing the genetic qualities of a population or breed through selection and mating systems. Some major strategies include:

1. **Selective Breeding**: This involves choosing animals with desirable traits (e.g., high milk yield, fast growth) as parents for the next generation. Over time, selective breeding increases the frequency of favorable genes in the population.
2. **Crossbreeding**: Mating individuals from different breeds to combine desirable traits from both parents. Crossbreeding can lead to **hybrid vigor (heterosis)**, where offspring outperform both parents in certain traits such as disease resistance or growth rate.
3. **Inbreeding**: Mating closely related animals to fix certain traits within a population. This is done cautiously to avoid inbreeding depression (reduced performance due to increased expression of harmful recessive genes).
4. **Artificial Insemination (AI)**: A reproductive technology that allows the use of semen from genetically superior sires across a wide population. AI helps spread superior genetics rapidly without moving animals physically.
5. **Genomic Selection and Biotechnology**: Advances in DNA testing and genome mapping allow breeders to identify animals with desirable genetic markers even before birth. Techniques such as marker-assisted selection (MAS), cloning, and gene editing (e.g., CRISPR) are increasingly used in modern breeding programs.

Examples of Breed Improvement

1. **Dairy Cattle (Holstein-Friesian)**: Selective breeding for high milk yield and improved udder conformation has made Holsteins the top dairy breed globally.

2. **Broiler Chickens**: Crossbreeding different lines of chickens with fast growth and good feed efficiency has led to the modern broiler that reaches market weight in 6–8 weeks.
3. **West African Dwarf Goats**: Crossbreeding with larger breeds like the Boer goat improves growth rate and meat production while maintaining disease resistance.
4. **N'Dama Cattle**: Known for trypanotolerance, N'Dama genetics are being used to improve disease resistance in other tropical cattle through crossbreeding.
5. **Sheep (Dorper x Local Breeds)**: Crossbreeding Dorper sheep with indigenous breeds improves carcass quality, growth rate, and adaptability to harsh environments.

CAUSES AND EFFECTS OF DISEASES IN LIVESTOCK

Livestock diseases are major constraints in animal production, reducing productivity, profitability, and animal welfare. These diseases are caused by various agents including **viruses**, **bacteria**, **fungi**, and **parasites** (both internal and external). Their effects range from mild symptoms to death, and can be transmitted through direct contact, vectors, contaminated feed, water, or poor environmental conditions.

1. Causes of Livestock Diseases

The common causes of diseases in livestock are:

- **Viral Agents**: Non-living infectious particles causing diseases such as Newcastle disease and Foot and Mouth Disease.
- **Bacterial Agents**: Microscopic organisms causing illnesses like Brucellosis, Anthrax, and Salmonellosis.
- **Fungal Agents**: Cause infections such as Aspergillosis and Ringworm, particularly in humid environments.
- **Parasitic Agents**: Include protozoa (e.g., coccidia), helminths (e.g., tapeworms), and ectoparasites (e.g., lice, mites).

2. Effects of Diseases on Livestock

- **Decreased productivity**: Lower meat, milk, and egg yields.
- **High mortality**: Especially in young and immunocompromised animals.
- **Reproductive failure**: Abortions, infertility, and neonatal deaths.
- **Economic loss**: High cost of treatment and control measures.
- **Public health risk**: Zoonotic diseases such as tuberculosis and brucellosis affect humans too.

3. Table of Common Diseases in Livestock

Animal	Disease	Causal Organism	Symptoms	Diagnosis	Prevention	Treatment & Management
Poultry	Newcastle Disease	Newcastle disease virus	Coughing, sneezing, green diarrhea, paralysis	Post-mortem, ELISA, PCR	Vaccination, biosecurity	No cure; supportive care & culling
	Coccidiosis	Eimeria spp. (protozoa)	Bloody droppings, ruffled feathers, lethargy	Fecal test	Sanitation, coccidiostats	Amprolium, sulfonamides
	Fowl Pox	Avian pox virus	Wart-like lesions, diphtheritic plaques	Clinical signs	Vaccination	Supportive therapy
	Salmonellosis	Salmonella spp. (bacteria)	Diarrhea, weakness, sudden death	Culture, serology	Hygiene, vaccination	Antibiotics, isolate infected
Cattle	Foot and Mouth Disease (FMD)	Aphthovirus	Drooling, lameness, mouth blisters	ELISA, PCR	Vaccination, quarantine	No cure; supportive care
	Brucellosis	Brucella abortus	Abortion, infertility	RBPT, ELISA	Vaccination (S19)	Culling, test and remove
	Mastitis	Staphylococcus, Streptococcus spp.	Swollen udder, milk clots	Milk test (CMT)	Milking hygiene	Antibiotics, proper milking
	Bovine Trypanosomiasis	Trypanosoma spp.	Emaciation, anemia	Blood smear	Tsetse fly control	Diminazene, isometamidium
Sheep & Goat	Peste des Petits Ruminants (PPR)	Morbillivirus	Fever, nasal discharge, diarrhea	PCR, ELISA	Vaccination	Supportive care, antibiotics for secondary infection

Animal	Disease	Causal Organism	Symptoms	Diagnosis	Prevention	Treatment & Management
	Foot Rot	Dichelobacter nodosus, Fusobacterium spp.	Lameness, foul-smelling hooves	Clinical exam	Footbath, hygiene	Antibiotics, hoof trimming
	Coccidiosis	Eimeria spp.	Diarrhea, weight loss	Fecal test	Sanitation	Coccidiostats
	Contagious Caprine Pleuropneumonia (CCPP)	Mycoplasma capricolum	Cough, labored breathing	Culture, serology	Vaccination	Antibiotics (oxytetracycline)
Rabbits	Snuffles	Pasteurella multocida	Nasal discharge, sneezing	Culture	Clean cages	Antibiotics (enrofloxacin)
	Coccidiosis	Eimeria spp.	Diarrhea, poor weight gain	Fecal exam	Clean environment	Sulfa drugs
	Ear Mites	Psoroptes cuniculi	Head shaking, crust in ears	Ear swab	Regular inspection	Miticides (ivermectin)
Snails	Shell Rot	Bacteria/Fungi	Shell erosion, odor	Visual inspection	Clean water, pH control	Disinfectants, salt baths
	Parasites	Nematodes, flukes	Poor growth, slime overproduction	Microscopy	Wash vegetables before feeding	Albendazole in small doses
	Edema Disease	Water imbalance/infection	Swelling, lethargy	Clinical signs	Avoid over-hydration	Adjust water, use salt solution
Pigs	African Swine Fever (ASF)	ASF virus	High fever, skin blotches, sudden death	PCR, ELISA	Biosecurity	No cure; culling infected pigs
	Swine Erysipelas	Erysipelothrix rhusiopathiae	Fever, skin lesions, arthritis	Culture	Vaccination	Penicillin, hygiene

Animal	Disease	Causal Organism	Symptoms	Diagnosis	Prevention	Treatment & Management
	Mange	Sarcoptes scabiei	Itching, crusty skin	Skin scraping	Clean pens	Ivermectin
	Worm Infestation	Ascaris suum	Poor growth, coughing	Fecal exam	Deworming schedule	Levamisole, albendazole

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4. General Prevention and Control Strategies

- **Biosecurity**: Limit access to farms, disinfect tools, and vehicles.
- **Vaccination**: A well-planned schedule for viral and bacterial diseases.
- **Hygiene and Sanitation**: Clean housing, feed, and water containers reduce exposure.
- **Regular Monitoring**: Early detection through clinical observation and diagnostic tests.
- **Proper Nutrition**: Enhances immune response and disease resistance.
- **Vector Control**: Use of insecticides and repellents to prevent disease transmission.